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RESEARCH ARTICLE

Multi-UAV Cooperative Suspension Control Strategy Considering Variable Rope Length and Load Posture Coupling Effect

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ABSTRACT This paper addresses the issues of insufficient model accuracy and poor control performance in traditional unmanned aerial vehicle (UAV) lifting operations by proposing an improved control strategy for multi-UAV cooperative lifting operations and optimizing the dynamic model. Firstly, the efficiency of the operation is enhanced, by introducing multi-UAV cooperative lifting control. On this basis, the traditional dynamic model is improved, with a focus on the impact of variable rope length, analyzing the dynamic coupling relationship between rope length and swing angle, and constructing related coupling error functions to improve the stability and control accuracy of the system. At the same time, the interaction between the load and the swing is studied, considering the velocity relationship between the suspension point and the load, and taking into account the coupled effect of the load attitude. Finally, a Lagrangian dynamic model that includes multiple-UAV, variable rope length, and load-coupled effects is established. Based on the improved model, this paper proposes a coupled error function, designs a nonlinear controller based on the Lyapunov method, which is non-passive, and realizes precise position tracking of the UAV and suppression of the load swing, while maintaining the rope length within the desired range. Finally, the stability and convergence of the closed-loop system are proven through Lyapunov stability analysis and the LaSalle invariance principle. The simulation results show that the proposed method exhibits significant advantages in UAVs' position, rope length tracking, and load swing suppression compared to the traditional lifting control strategy, and has strong robustness against uncertain parameters and external disturbances, verifying the effectiveness and practical value of the proposed method.

INDEX TERMS Cooperative control, multi-UAV lifting, nonlinear control, nonlinear coupled system, swing control, time-varying rope length.

I. INTRODUCTION

The growing demand for aerial transportation in complex environments, such as disaster relief operations requiring payload delivery in unstructured terrains, construction tasks

in urban canyons, or military logistics in hostile areas has created an urgent need for robust multi-UAV cooperative lifting systems [1], [2], [3]. While existing studies have advanced single-UAV lifting control, real-world industrial and emergency scenarios often require collaborative fleets to handle heavy payloads that exceed the capacity of individual drones. Furthermore, practical operations inevitably involve dynamic

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adjustments of cable lengths and load attitude variations during mid-air reorientation. And these factors critically absent in conventional fixed-cable models [4], [5].

For the implementation of UAV lifting, the main method is to hang the load through a rope and fix it to the UAV, creating an application scheme for the UAV lifting system. The UAV lifting system is a strongly coupled, nonlinear, and multi-variable physical system. During the lifting process, the load may experience swinging, rotation, and even complex motion patterns with multiple degrees of freedom, which directly affect the stability and controllability of the UAV. At the same time, the uncertainty in the load's mass and external disturbances also pose significant challenges for the design of the controller. Current research primarily focuses on the dynamic modeling and control of the system [6]. There are significant limitations in the dynamic modeling research of quadrotor UAV lifting. Long et al. proposed an event-triggered mechanism under full-state constraints, which reduces the frequency of controller updates [7]. However, its fixed constraint boundary assumption cannot adapt to the dynamic adjustment requirements of the rope length. Chang et al. introduced a rope with elasticity and damping, reducing the swing of the load by increasing the system's dissipative properties [8]. Alattas et al. developed an obstacle function compensation strategy [9], which addresses the issue of input saturation but overlooks the coupling effects of three-dimensional attitude. Yang et al. proposed a load swing angle dynamics model that includes a variable-length cable related to the acceleration of the quadrotor flight [10]. Although it introduced the factor of rope length variation, it can not treat it as an independent control variable, thereby limiting the capability for active swing suppression. The establishment of an accurate model is the foundation for ensuring system stability, but the efficient lifting also closely relates to the design of the controller. Compared to the flight control of independent quadrotor drones, the flight control problem of drone load transportation is more complex. However, existing solutions have common defects. Although the adaptive sliding mode strategy by Najafi et al. eliminated chattering through the hyperbolic tangent function [11], its reliance on the upper bound of disturbances will lead to a significant increase in tracking error in scenarios where the rope length changes suddenly. Kuszniir and Smoczek combined a sliding mode controller with feedback linearization to minimize the vibration of the load [12]. Zhou et al. proposed an anomaly detection scheme based on wavelet decomposition and deep feature extraction, which improved perception accuracy in noisy environments [13], but the processing delay may not meet real-time control requirements. Santos et al. proposed a strategy based on a non-sampled Kalman filter (UKF) and model predictive control (MPC) to achieve stable trajectory tracking for the lifting system under parameter uncertainty [14]. Fan et al. introduced a neural network adaptive control algorithm to compensate for any external disturbances [15]. However, they all face the computational burden of online

optimization, making it difficult to achieve multivariable dynamic coordination. Other methods such as LQR-PID control, geometric nonlinear control, cascaded control, feedback linearization control, robust control, and passive control also have similar limitations, urgently requiring breakthroughs in new control architectures [16], [17], [18], [19], [20], [21].

The above research primarily focused on a single UAV as the control object. However, the payload capacity of a single UAV is limited. Bernard and Kondak demonstrated the first use of three helicopters for lifting and transporting a load [22], based on a small unmanned helicopter payload transportation system. Subsequently, many scholars have conducted research in related areas; however, to date, multi-machine coordination schemes still face key bottlenecks. Chopra and Ghose adopted the dynamic coupling relationship between the translational motion of the UAVs and the rotational motion of the payload for vertical formation control [23]. However, the model only addressed vertical dynamics coupling, neglecting horizontal dynamic correlations. Yin et al. proposed a single-machine navigation strategy based on deep reinforcement learning [24], but it cannot handle multi-machine dynamics coupling, due to its isolated architecture. Villa et al. proposed the method connecting two UAVs and the payload through a flexible rope and controlling them as a single system. The virtual structure method was used to maintain the formation, which was a feasible solution to enhance the payload capacity [25]. The model can not involve the complete three-dimensional space, resulting in a lack of precision in practical applications. Zhang et al. addressed the formation planning problem for tethered multirotor UAV cooperative transportation with unknown payload and cable lengths, designing an online formation tracking control based on estimation [26]. Although rope length estimation was introduced, a closed-loop feedback mechanism was not established. In the process of establishing the model, the influence of the load attitude coupling on the swing control of the lifting was ignored, leading to a reduction in the accuracy of the system. Furthermore, most existing research assumes that the cable length remains constant. Pizetta et al. proposed a space obstacle avoidance tracking algorithm that ensures the stable cooperative lifting of two UAVs while avoiding the risk of mutual collision and collision with obstacles [27]. However, although the model considered the cable to be an elastic rope, the original length of the cable was still assumed to be a fixed constant, not treated as an independent variable. The fixed original length assumption limited the potential for active swing suppression. In practical applications, changes in the length of the cable can also suppress the swing of the payload to some extent.

In summary, the problem of swing control for a quadrotor UAV lifting system, considering changes in rope length and the rotation of the suspended load, remains an issue that requires further research. Existing research has not fully considered the simultaneous impact of variable rope length, load attitude coupling, multi-UAV positioning, and swing

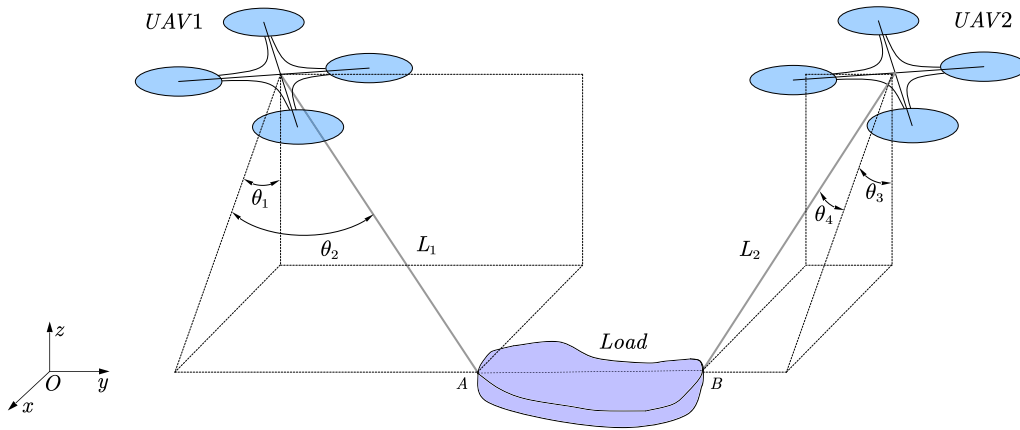


FIGURE 1. Physical model of Dual-UAV cooperative lifting.

suppression, resulting in a reduction in model accuracy. This not only affects the actual control effect but also reduces the practicality of the strategy. In addition, among the control methods for this problem, many are feedforward controls, while feedback control methods have not yet incorporated changes in rope length and load attitude coupling into the multi-UAV positioning control and load swing suppression.

To address the aforementioned issues and achieve more precise control of the oscillation of multiple UAV loads, this paper proposes a systematic innovative solution: (1) Fully coupled dynamic modeling: overcoming the fixed rope length assumption by innovatively introducing winch driving force as a fourth control dimension, constructing a complete model with dynamic rope length adjustment, load attitude coupling, and multi-UAV coordinated positioning; (2) Collaborative control architecture design: designing analytical auxiliary error functions to explicitly express the dynamic relationship among displacement, rope length, and swing angle; (3) Passive control strategy: designing a nonlinear controller based on Lyapunov stability theory to simultaneously handle input saturation and state constraints through the principle of energy dissipation.

The rest of the paper is organized as follows: Section II introduces comprehensive dynamic modeling of the multi-UAV cooperative lifting system, considering the variable-length rope effect and load attitude coupling. Section III elaborates on the design of auxiliary error functions and establishes the dynamic coupling relationships between UAV displacement, rope length, and swing angle dynamics. Section IV systematically designs a nonlinear control algorithm based on Lyapunov stability theory through the passive energy dissipation principle, providing a rigorous mathematical proof of the system's stability and convergence characteristics. Section V offers extensive simulation results to validate the controller's performance across various operational scenarios, including robustness tests against parameter uncertainties and external disturbances. Finally, Section VI

summarizes the main contributions of the paper and discusses potential future research directions.

II. DYNAMIC MODEL

Multi-UAV cooperative lifting typically involves two or more unmanned aerial vehicles (UAVs), each connected to the load through a rope, to achieve cargo transportation. During the transportation process, it is necessary not only to ensure that the UAVs accurately reach their destinations but also to minimize the swing of the load as much as possible. Fig. 1 shows the physical model of a Dual-UAV cooperative lifting system.

Define an inertial coordinate system $O - xyz$ fixed on the ground, as shown in Fig. 1. The position vectors of UAV 1 and UAV 2 are $\mathbf{p}_1 = (x_1, y_1, z_1)^T$, $\mathbf{p}_2 = (x_2, y_2, z_2)^T$, respectively. The position vector of the load is $\mathbf{p}_3 = (x_3, y_3, z_3)^T$. The length vectors of the ropes 1 and 2 are $\mathbf{L} = (L_1, L_2)^T$, which can change with time t . Define the swing angles of the ropes 1 and 2 with respect to the plane XOZ and the plane YOZ as $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3, \theta_4]^T$.

In the lifting system model proposed in this paper, as shown in Fig. 1, it is considered that the UAVs can move freely in the x, y, z directions. Due to the presence of the ropes, the load will also experience position displacement and swing as the UAVs move.

To facilitate a more targeted study of the multi-UAV cooperative lifting process, the following assumptions are made:

1. The UAVs are considered as point masses, ignoring the extreme case of UAV-UAV collisions;
2. The load is treated as a rigid body with uniform mass distribution, and the suspension point is symmetrical about the center of mass;
3. The mass of the ropes connecting the UAVs and the load is ignored, and the ropes are assumed to be always taut;
4. The range of the load's swing angle is $-\frac{\pi}{2} < \theta_1, \theta_2, \theta_3, \theta_4 < \frac{\pi}{2}$.

During the multi-UAV lifting process, the swing angle is primarily generated by the relative motion between the UAVs and the load. Changes in the length of the ropes and the attitude changes of the load itself also affect the magnitude of the swing angle.

The dynamic rope length generalized degrees of freedom are introduced into the dynamic model, and the coupling relationship between the UAVs and the load is established based on the dynamic rope length.

The position of the load's center of mass can be represented as:

$$\begin{aligned} x_3 &= (x_1 + L_1 \sin \theta_1 \cos \theta_2 + x_2 + L_2 \sin \theta_3 \cos \theta_4)/2 \\ y_3 &= (y_1 + L_1 \sin \theta_2 + y_2 + L_2 \sin \theta_4)/2 \\ z_3 &= (z_1 - L_1 \cos \theta_1 \cos \theta_2 + z_2 - L_2 \cos \theta_3 \cos \theta_4)/2 \end{aligned} \quad (1)$$

The load's attitude coupling is introduced into the model. Assuming that the load's two ends are suspended at points A and B, their respective positions can be represented as:

$$\begin{aligned} p_A &= [x_1 + L_1 \sin \theta_1 \cos \theta_2, y_1 + L_1 \sin \theta_2, z_1 \\ &\quad - L_1 \cos \theta_1 \cos \theta_2]^T \\ p_B &= [x_2 + L_2 \sin \theta_3 \cos \theta_4, y_2 + L_2 \sin \theta_4, z_2 \\ &\quad - L_2 \cos \theta_3 \cos \theta_4]^T \end{aligned} \quad (2)$$

And it satisfies $\|p_A - p_B\| = l$, where l is the distance between the suspension points.

The rotation angular velocity of the load is related to the velocity of the suspension points, with the specific relationship being:

$$\begin{aligned} \dot{p}_A &= \dot{p}_3 + \omega_3 \times r_A \\ \dot{p}_B &= \dot{p}_3 + \omega_3 \times r_B \end{aligned} \quad (3)$$

where ω_3 is the rotation angular velocity vector of the load, and r_A, r_B is the position vector of suspension points A, B with respect to the load's center of mass.

After transformation, the angular velocity vector of the load can be represented as:

$$\omega_3 = \frac{(r_B - r_A) \times (\dot{p}_B - \dot{p}_A)}{|r_B - r_A|^2} \quad (4)$$

The system's dynamic model is established based on Lagrangian mechanics. The kinetic energy E_k of the multi-UAV cooperative lifting system includes both translational and rotational kinetic energies, as follows:

$$E_k = \frac{1}{2} m_1 \dot{p}_1^T \dot{p}_1 + \frac{1}{2} m_2 \dot{p}_2^T \dot{p}_2 + \frac{1}{2} m_3 \dot{p}_3^T \dot{p}_3 + \frac{1}{2} \omega_3^T I_3 \omega_3 \quad (5)$$

where m_1 and m_2 are the masses of UAV 1 and UAV 2, respectively, m_3 is the mass of the load, and $I_3 = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ is the inertia tensor matrix of the load with respect to its body coordinate system.

The potential energy E_p of the entire system is:

$$E_p = m_1 g z_1 + m_2 g z_2 + m_3 g z_3 \quad (6)$$

In which g is the acceleration due to gravity.

The resulting Lagrangian function is expressed as:

$$\begin{aligned} L &= E_k - E_p \\ &= \left(\frac{1}{2} m_1 \dot{p}_1^T \dot{p}_1 + \frac{1}{2} m_2 \dot{p}_2^T \dot{p}_2 + \frac{1}{2} m_3 \dot{p}_3^T \dot{p}_3 + \frac{1}{2} \omega_3^T I_3 \omega_3 \right) \\ &\quad - (m_1 g z_1 + m_2 g z_2 + m_3 g z_3) \end{aligned} \quad (7)$$

The system's dynamic model is in the standard form as follows:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = U \quad (8)$$

where $M(q)$ is the inertia matrix, $C(q, \dot{q})$ is the centrifugal-Coriolis matrix, $G(q)$ is the gravity matrix, and U is the generalized force matrix. q is the generalized coordinate, with the specific form being as follows:

$$q = [x_1, y_1, z_1, x_2, y_2, z_2, L_1, L_2, \theta_1, \theta_2, \theta_3, \theta_4]^T \quad (9)$$

The matrix U introduces the components of the thrust of UAV 1,2 in the X, Y, Z directions of the inertial coordinate system, as well as the driving force of the hoist for controlling the winding and unwinding of ropes 1,2 as generalized forces. It can be represented as:

$$U = u = [u_1, u_2, u_3, u_4, u_5, u_6, u_7, u_8, 0_{1 \times 4}]^T \quad (10)$$

The dynamic model contains a large number of coupled terms, and the coupled terms have highly nonlinear characteristics. When the swing angle, rope length, and load attitude change dramatically, it will have a significant impact on the flight of the UAVs. Additionally, the dynamic changes in the swing angle are nonlinearly coupled with the load attitude, the acceleration of the UAVs, and the time derivative of the rope length. Considering that the external inputs to the entire system are only provided by the UAV's thrust and the driving force of the hoist, the above information indicates that a controller needs to be designed to appropriately adjust the UAV's flight thrust and the length of the rope to suppress the swing of the cable.

III. DESIGN OF A PASSIVE CONTROLLER BASED ON ENERGY ANALYSIS

The displacements of the UAVs in the X, Y, Z directions are directly controlled by the UAV's thrust, and the length of the ropes is directly controlled by the driving force of the hoist. The swing angles corresponding to the two groups of ropes connected to UAV 1 and UAV 2 are primarily determined by the relative motion between the UAVs and the load. They are indirectly controlled by the UAV's motion in the X, Y, Z directions and the length of the rope. Thus, the swing angles are indirectly controlled by the UAV's thrust and the driving force of the hoist. The total degrees of freedom of the system are 12, and the number of control parameters is 8. The number of degrees of freedom for the load's swing angle and displacement variables is greater than the total number of control input variables for the UAVs, indicating that this system is a typical nonlinear underactuated system. Currently, the design of controllers for such systems is still not complete.

The method of passive control based on energy analysis is widely applied in the nonlinear feedback control design of UAV lifting systems. In this paper, energy analysis is performed on the multi-UAV cooperative lifting system, and an auxiliary error function related to the swing angle is proposed. Based on the system's energy storage function, a Lyapunov candidate function is established, and the controller is designed accordingly.

For the underactuated cooperative lifting system proposed in this paper, the control goal is to ensure that the UAVs accurately and quickly reach the desired positions while suppressing the swing of the load and maintaining the length of the ropes unchanged. The system's desired objectives are as follows:

$$\lim_{t \rightarrow \infty} \mathbf{q} = \mathbf{q}_d \quad (11)$$

where $\mathbf{q}_d = [x_{1d}, y_{1d}, z_{1d}, x_{2d}, y_{2d}, z_{2d}, L_{1d}, L_{2d}, \theta_{1d}, \theta_{2d}, \theta_{3d}, \theta_{4d}]^T \in \mathbb{R}^{12}$ contains the desired control objectives for the UAV positions, rope lengths, and swing angles.

The tracking error is defined as:

$$\begin{aligned} e_{x1} &= x_1 - x_{1d}, e_{y1} = y_1 - y_{1d}, e_{z1} = z_1 - z_{1d}, e_{L1} \\ &= L_1 - L_{1d} \\ e_{x2} &= x_2 - x_{2d}, e_{y2} = y_2 - y_{2d}, e_{z2} = z_2 - z_{2d}, e_{L2} \\ &= L_2 - L_{2d} \end{aligned} \quad (12)$$

The control forces of the UAVs and the hoist only act on the displacements of the UAVs and the lengths of the ropes, respectively, and cannot directly affect the swinging angle. Therefore, it is not possible to directly use the forces to control the swing angle. Using the aforementioned error as a feedback signal, the control forces can be used for the flight positioning of the lifting system, but the control effect on the swinging angle is poor. Existing control methods may not be able to effectively suppress the swing of the load, which is a common challenge for many underactuated systems.

In order to achieve the suppression of swinging oscillations while ensuring the flight positioning of the UAVs, a driving relationship between the control forces and the swinging angle is established. Based on the improved model, the driving force u_7, u_8 of the hoist for controlling the rope length is introduced as an input to enhance the control effect, and the auxiliary error function is used as a feedback input, providing dynamic coupling between the positions, rope lengths, and swinging angles, thereby achieving the expected control objectives. The definition of the auxiliary error function is as follows:

$$\begin{aligned} \varepsilon_{x1} &= e_{x1} - f_{x1}, \varepsilon_{y1} = e_{y1} - f_{y1}, \varepsilon_{z1} = e_{z1} - L_{1d} \\ &\quad + f_{z1}, \varepsilon_{L1} = e_{L1} \\ \varepsilon_{x2} &= e_{x2} - f_{x2}, \varepsilon_{y2} = e_{y2} - f_{y2}, \varepsilon_{z2} = e_{z2} - L_{2d} \\ &\quad + f_{z2}, \varepsilon_{L2} = e_{L2} \end{aligned} \quad (13)$$

where $f_{x1} = L_1 \sin \theta_1 \cos \theta_2$; $f_{y1} = L_1 \sin \theta_2$; $f_{z1} = L_1 \cos \theta_1 \cos \theta_2$; $f_{x2} = L_2 \sin \theta_3 \cos \theta_4$; $f_{y2} = L_2 \sin \theta_4$; $f_{z2} = L_2 \cos \theta_3 \cos \theta_4$.

To facilitate subsequent analysis, the following is defined:

$$\begin{aligned} \mathbf{e} &= [e_{x1}, e_{y1}, e_{z1}, e_{x2}, e_{y2}, e_{z2}, e_{L1}, e_{L2}]^T \\ \boldsymbol{\varepsilon} &= [\varepsilon_{x1}, \varepsilon_{y1}, \varepsilon_{z1}, \varepsilon_{x2}, \varepsilon_{y2}, \varepsilon_{z2}, \varepsilon_{L1}, \varepsilon_{L2}]^T \\ \mathbf{f} &= [f_{x1}, f_{y1}, f_{z1}, f_{x2}, f_{y2}, f_{z2}]^T \end{aligned} \quad (14)$$

When the system's swing is controlled, $\mathbf{f} \rightarrow \mathbf{0}_{6 \times 1}$, and thus (13) can be simplified to (12), indicating that when the load's swing is suppressed, the UAVs can better track the desired path.

Through an energy analysis of the underactuated cooperative lifting system, the system's energy storage function is established, and on this basis, the auxiliary error function is used to construct the following Lyapunov candidate function:

$$V_1 = E(t) + E_\varepsilon \quad (15)$$

where $E(t)$ is the total energy of the system. $E(t)$ is defined as follows:

$$E(t) = E_k + E_p \quad (16)$$

By taking the time derivative of $E(t)$ and applying the principle of energy, we obtain:

$$\begin{aligned} \dot{E}(t) &= u_1 \dot{x}_1 + u_2 \dot{y}_1 + u_3 \dot{z}_1 + u_4 \dot{x}_2 + u_5 \dot{y}_2 \\ &\quad + u_6 \dot{z}_2 + u_7 \dot{L}_1 + u_8 \dot{L}_2 \end{aligned} \quad (17)$$

The function E_ε is defined as:

$$E_\varepsilon = \frac{1}{2} \sum_{i=1}^8 k_{p1}^i \varepsilon_i^2 \quad (18)$$

In the above equation, k_{p1}^i is a positive gain coefficient, and it can be seen that E_ε is a non-negative function.

Let $\varepsilon_i = \varepsilon_{x1}, \varepsilon_{y1}, \varepsilon_{z1}, \varepsilon_{x2}, \varepsilon_{y2}, \varepsilon_{z2}, \varepsilon_{L1}, \varepsilon_{L2}$ ($i = 1, 2, \dots, 8$), and when $\varepsilon_i \rightarrow 0$, take the minimum value. Under the assumption that the ropes are always taut, it can be seen that $E(t)$ is a positive definite function. It can be deduced that $V_1 \geq 0$. Additionally, the E_ε function introduces nonlinear coupling between the state variables. Theoretically, the controller can perform two tasks simultaneously, namely, completing the tracking task while suppressing the swing of the payload. However, although E_ε can introduce coupling between the state variables, it cannot fully compensate for the coupling of each variable solely by V_1 . Therefore, to achieve better control coupling, a new Lyapunov candidate function is needed:

$$V_2 = \sum_{i=1}^8 \left(\frac{1}{2} [k_{p2}^i \varepsilon_i \arctan \varepsilon_i + \frac{1}{2} k_{p3}^i \ln(1 + \varepsilon_i^2)] \right) \quad (19)$$

In the equation, k_{p2}^i, k_{p3}^i is a positive gain coefficient. Similarly, it can be seen that V_2 is a non-negative function, taking its minimum value when $\varepsilon_i \rightarrow 0$.

Thus, the final Lyapunov candidate function is:

$$V(t) = V_1 + V_2 \geq 0 \quad (20)$$

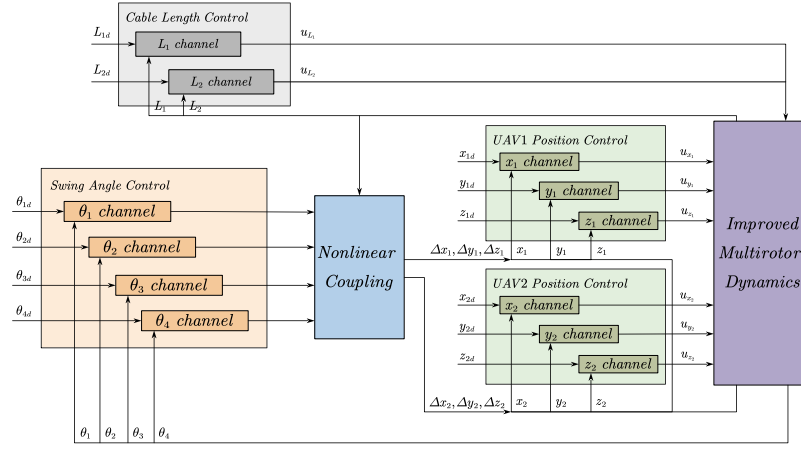


FIGURE 2. Tracking and swing control framework for Dual-UAV lifting based on passive analysis.

The time derivative of $V(t)$ is:

$$\dot{V}(t) = \dot{E}(t) + \sum_{i=1}^8 (k_{p1}^i \varepsilon_i \dot{\varepsilon}_i + k_{p2}^i \dot{\varepsilon}_i \arctan \varepsilon_i + k_{p3}^i \frac{\varepsilon_i \dot{\varepsilon}_i}{1 + \varepsilon_i^2}) \quad (21)$$

Based on the Lyapunov principle, the feedback controller for UAV positioning, rope length tracking, and swing suppression is constructed as follows:

$$u_i = -k_{p1}^i \varepsilon_i - k_{d1}^i \dot{\varepsilon}_i - \frac{1}{2} k_{p2}^i \arctan \varepsilon_i - k_{d2}^i \arctan \dot{\varepsilon}_i - k_{p3}^i \frac{\varepsilon_i}{1 + \varepsilon_i^2} - k_{d3}^i \frac{\dot{\varepsilon}_i}{1 + \dot{\varepsilon}_i^2} \quad (i = 1, 2, 3, \dots, 8) \quad (22)$$

In the equation, $k_{d1}^i, k_{d2}^i, k_{d3}^i$ is a positive gain coefficient. Let $k_i = [k_{p1}^i, k_{d1}^i, k_{p2}^i, k_{d2}^i, k_{p3}^i, k_{d3}^i]^T$.

The gain coefficient matrix is defined as:

$$K = [k_1, k_2, k_3, k_4, k_5, k_6, k_7, k_8]^T \quad (23)$$

The controller establishes a coupling control relationship through the gain coefficient matrix. Substituting the above controller into (8) reveals that the equation introduces additional nonlinear coupling between the state variables. The overall design framework of the controller is shown in Fig. 2.

IV. STABILITY AND CONVERGENCE ANALYSIS

A. STABILITY ANALYSIS

The stability of the designed passive nonlinear controller in the closed-loop system is verified. The time derivative of $V(t)$ is obtained as follows $\dot{V}(t)$:

$$\dot{V}(t) = \dot{V}_1 + \dot{V}_2 \quad (24)$$

where:

$$\dot{V}_1 = \sum_{i=1}^8 u_i \dot{\varepsilon}_i + \sum_{i=1}^8 k_{p1}^i \varepsilon_i \dot{\varepsilon}_i \quad (25)$$

The time derivative of V_2 is:

$$\dot{V}_2 = \sum_{i=1}^8 (k_{p2}^i \dot{\varepsilon}_i \arctan \varepsilon_i + k_{p3}^i \frac{\varepsilon_i \dot{\varepsilon}_i}{1 + \varepsilon_i^2}) \quad (26)$$

Substituting (25) and (26) into (24) yields:

$$\dot{V}(t) = \sum_{i=1}^8 u_i \dot{\varepsilon}_i + \sum_{i=1}^8 (k_{p1}^i \varepsilon_i \dot{\varepsilon}_i + k_{p2}^i \dot{\varepsilon}_i \arctan \varepsilon_i + k_{p3}^i \frac{\varepsilon_i \dot{\varepsilon}_i}{1 + \varepsilon_i^2}) \quad (27)$$

After simplification, further yields:

$$\dot{V}(t) = \sum_{i=1}^8 (k_{p1}^i \varepsilon_i + \frac{1}{2} k_{p2}^i \arctan \varepsilon_i + k_{p3}^i \frac{\varepsilon_i}{1 + \varepsilon_i^2} + u_i) \dot{\varepsilon}_i \quad (28)$$

Introducing the expression of the controller (22), it can be derived that:

$$\dot{V}(t) = \sum_{i=1}^8 (-k_{d1}^i \dot{\varepsilon}_i^2 - k_{d2}^i \dot{\varepsilon}_i \arctan \dot{\varepsilon}_i - k_{d3}^i \frac{\dot{\varepsilon}_i^2}{1 + \dot{\varepsilon}_i^2}) \leq 0 \quad (29)$$

Therefore, the closed-loop multi-UAV cooperative lifting system under the given controller is asymptotically stable according to Lyapunov theory.

1) CONVERGENCE ANALYSIS

Theorem 1: For the proposed passive control strategy, the multi-UAV lifting system can be closed-loop stable and the state can converge asymptotically to the desired position:

$$\lim_{t \rightarrow \infty} \mathbf{q} = \mathbf{q}_d, \lim_{t \rightarrow \infty} \dot{\mathbf{q}} = \mathbf{0}_{12 \times 1} \quad (30)$$

Proof: Since the system is asymptotically stable, as $\dot{V}(t) \leq 0$, then $V(t) \in \mathcal{L}_\infty$. Where $\mathcal{L}_\infty$ represents a

bounded function set. From (20) and (21), the conclusion can be obtained:

$$\mathbf{q}, \dot{\mathbf{q}}, \mathbf{f}, \dot{\mathbf{f}} \in \mathcal{L}_{\infty} \quad (31)$$

Analysis based on (12), (13), and (24) yields:

$$\mathbf{e}, \dot{\mathbf{e}}, \ddot{\mathbf{e}}, \mathbf{u} \in \mathcal{L}_{\infty} \quad (32)$$

Next, the proof is completed using LaSalle invariance principle, defining the following invariant set:

$$\xi = \{(\mathbf{q}, \dot{\mathbf{q}}) | \dot{V}(t) = 0\} \quad (33)$$

The set ζ is the largest invariant subset of the invariant set ξ , and the set ζ contains the following equilibrium points:

$$\dot{\mathbf{e}} = \mathbf{0}_{8 \times 1}, \ddot{\mathbf{e}} = \mathbf{0}_{8 \times 1} \quad (34)$$

Thus, it can be concluded that:

$$\begin{aligned} \dot{\mathbf{p}}_1 &= \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \\ \dot{z}_1 \end{bmatrix} = \begin{bmatrix} \dot{f}_{x1} \\ \dot{f}_{y1} \\ -\dot{f}_{z1} \end{bmatrix}, \dot{\mathbf{p}}_2 = \begin{bmatrix} \dot{x}_2 \\ \dot{y}_2 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} \dot{f}_{x2} \\ \dot{f}_{y2} \\ -\dot{f}_{z2} \end{bmatrix}, \dot{\mathbf{L}} = \begin{bmatrix} \dot{L}_1 \\ \dot{L}_2 \end{bmatrix} \\ &= \mathbf{0}_{2 \times 1} \\ \ddot{\mathbf{p}}_1 &= \begin{bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \\ \ddot{z}_1 \end{bmatrix} = \begin{bmatrix} \ddot{f}_{x1} \\ \ddot{f}_{y1} \\ -\ddot{f}_{z1} \end{bmatrix}, \ddot{\mathbf{p}}_2 = \begin{bmatrix} \ddot{x}_2 \\ \ddot{y}_2 \\ \ddot{z}_2 \end{bmatrix} = \begin{bmatrix} \ddot{f}_{x2} \\ \ddot{f}_{y2} \\ -\ddot{f}_{z2} \end{bmatrix}, \ddot{\mathbf{L}} = \begin{bmatrix} \ddot{L}_1 \\ \ddot{L}_2 \end{bmatrix} \\ &= \mathbf{0}_{2 \times 1} \end{aligned} \quad (35)$$

Assume that $\mathbf{e} = [a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8]^T \neq \mathbf{0}_{8 \times 1}$, and $\mathbf{e} \in \mathbb{R}^8$.

Substituting (34) into the controller (22) yields:

$$u_i = -k_{p1}^i \varepsilon_i - \frac{1}{2} k_{p2}^i \arctan \varepsilon_i - k_{p3}^i \frac{\varepsilon_i}{1 + \varepsilon_i^2} \quad (36)$$

Substituting (35) and (36) into (8) yields:

$$\begin{aligned} \ddot{x}_1 &= \frac{1}{m_1} (-k_{p1}^1 a_1 - \frac{1}{2} k_{p2}^1 \arctan a_1 - k_{p3}^1 \frac{a_1}{1 + a_1^2}) \\ \ddot{y}_1 &= \frac{1}{m_1} (-k_{p1}^2 a_2 - \frac{1}{2} k_{p2}^2 \arctan a_2 - k_{p3}^2 \frac{a_2}{1 + a_2^2}) \\ \ddot{z}_1 &= \frac{1}{m_1} (-k_{p1}^3 a_3 - \frac{1}{2} k_{p2}^3 \arctan a_3 - k_{p3}^3 \frac{a_3}{1 + a_3^2}) \\ \ddot{x}_2 &= \frac{1}{m_2} (-k_{p1}^4 a_4 - \frac{1}{2} k_{p2}^4 \arctan a_4 - k_{p3}^4 \frac{a_4}{1 + a_4^2}) \\ \ddot{y}_2 &= \frac{1}{m_2} (-k_{p1}^5 a_5 - \frac{1}{2} k_{p2}^5 \arctan a_5 - k_{p3}^5 \frac{a_5}{1 + a_5^2}) \\ \ddot{z}_2 &= \frac{1}{m_2} (-k_{p1}^6 a_6 - \frac{1}{2} k_{p2}^6 \arctan a_6 - k_{p3}^6 \frac{a_6}{1 + a_6^2}) \\ \ddot{L}_1 &= \frac{2}{m_3} (-k_{p1}^7 a_7 - \frac{1}{2} k_{p2}^7 \arctan a_7 - k_{p3}^7 \frac{a_7}{1 + a_7^2}) \\ \ddot{L}_2 &= \frac{2}{m_3} (-k_{p1}^8 a_8 - \frac{1}{2} k_{p2}^8 \arctan a_8 - k_{p3}^8 \frac{a_8}{1 + a_8^2}) \end{aligned} \quad (37)$$

Assuming that $\mathbf{e} \neq \mathbf{0}_{8 \times 1}$ holds at this time, based on the above equation, it can be concluded that:

$$\lim_{t \rightarrow \infty} (\dot{x}_1 = \dot{x}_1(0) + \ddot{x}_1 \cdot t) = \infty, \lim_{t \rightarrow \infty} (\dot{y}_1 = \dot{y}_1(0) + \ddot{y}_1 \cdot t)$$

TABLE 1. Parameters used during the simulation.

Parameter	Description	Value/Units
m_1	Mass of the UAV 1	2 kg
m_2	Mass of the UAV 2	2 kg
m_3	Mass of the load	0.2 kg
g	Gravitational constant	9.8 m/s ²
I_{xx}	Inertia in the x-axis	0.1 kg·m ²
I_{yy}	Inertia in the y-axis	0.1 kg·m ²
I_{zz}	Inertia in the z-axis	0.1 kg·m ²
L	Length of the cable	(1 m, 1 m)
θ	Angle of the swing	(0, 0, 0, 0)
l	Distance of suspension points	0.5 m
p_1	Initial position of the UAV 1	(1 m, 1 m, 1 m)
p_2	Initial position of the UAV 2	(1 m, 1.5 m, 1 m)
p_{1d}	Desired position of the UAV 1	(4 m, 2 m, 3 m)
p_{2d}	Desired position of the UAV 2	(4 m, 2.5 m, 3 m)
L_d	Desired length of the cable	(1.5 m, 1.5 m)
θ_d	Desired angle of the swing	(0, 0, 0, 0)

$$\begin{aligned} &= \infty, \\ \lim_{t \rightarrow \infty} (\dot{z}_1 = \dot{z}_1(0) + \ddot{z}_1 \cdot t) &= \infty, \lim_{t \rightarrow \infty} (\dot{L}_1 = \dot{L}_1(0) + \ddot{L}_1 \cdot t) \\ &= \infty, \\ \lim_{t \rightarrow \infty} (\dot{x}_2 = \dot{x}_2(0) + \ddot{x}_2 \cdot t) &= \infty, \lim_{t \rightarrow \infty} (\dot{y}_2 = \dot{y}_2(0) + \ddot{y}_2 \cdot t) \\ &= \infty, \\ \lim_{t \rightarrow \infty} (\dot{z}_2 = \dot{z}_2(0) + \ddot{z}_2 \cdot t) &= \infty, \lim_{t \rightarrow \infty} (\dot{L}_2 = \dot{L}_2(0) + \ddot{L}_2 \cdot t) \\ &= \infty \end{aligned} \quad (38)$$

This is clearly contradictory to (31) and (32), therefore the initial assumption $\mathbf{e} \neq \mathbf{0}_{8 \times 1}$ does not hold, $\mathbf{e} = \mathbf{0}_{8 \times 1}$. Therefore, the following conclusion is drawn:

$$\begin{aligned} \mathbf{e} &= \mathbf{0}_{8 \times 1}, \mathbf{e} = \mathbf{0}_{8 \times 1}, \dot{\mathbf{f}} = \mathbf{0}_{6 \times 1}, \ddot{\mathbf{f}} = \mathbf{0}_{6 \times 1}, \mathbf{u} = \mathbf{0}_{8 \times 1} \\ \dot{x}_1 &= \dot{y}_1 = \dot{z}_1 = \dot{x}_2 = \dot{y}_2 = \dot{z}_2 = \dot{L}_1 = \dot{L}_2 = 0 \\ \ddot{x}_1 &= \ddot{y}_1 = \ddot{z}_1 = \ddot{x}_2 = \ddot{y}_2 = \ddot{z}_2 = \ddot{L}_1 = \ddot{L}_2 = 0 \end{aligned} \quad (39)$$

Based on (39) and taking the derivative of (14), given the model assumption $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, it can be solved that:

$$\begin{aligned} \theta_1 &= \theta_2 = \theta_3 = \theta_4 = 0 \\ \dot{\theta}_1 &= \dot{\theta}_2 = \dot{\theta}_3 = \dot{\theta}_4 = 0 \end{aligned} \quad (40)$$

In summary, the conclusions derived from (39) and (40) indicate that:

$$\lim_{t \rightarrow \infty} \mathbf{q} = \mathbf{q}_d, \lim_{t \rightarrow \infty} \dot{\mathbf{q}} = \mathbf{0}_{12 \times 1} \quad (41)$$

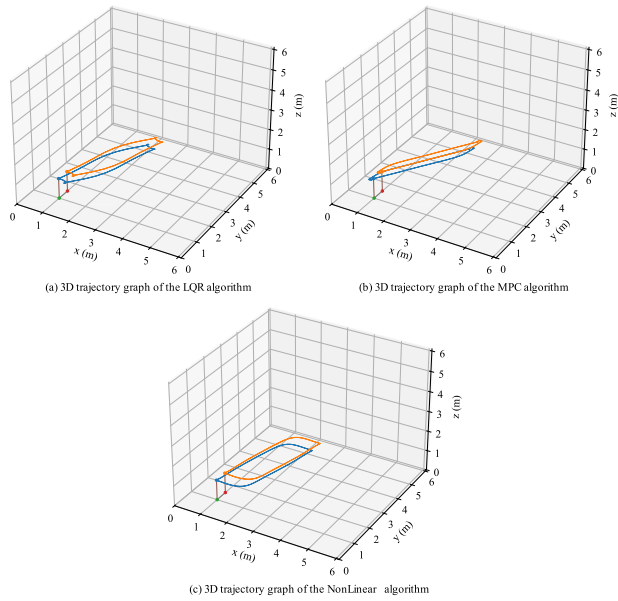


FIGURE 3. 3D Trajectory of UAVs lifting operation.

The largest invariant set ξ contains the desired equilibrium point $[q; \dot{q}] = [q_d; \dot{q}_d]$. According to LaSalle invariance principle, Theorem 1 is thus proven.

V. NUMERICAL SIMULATION RESULTS

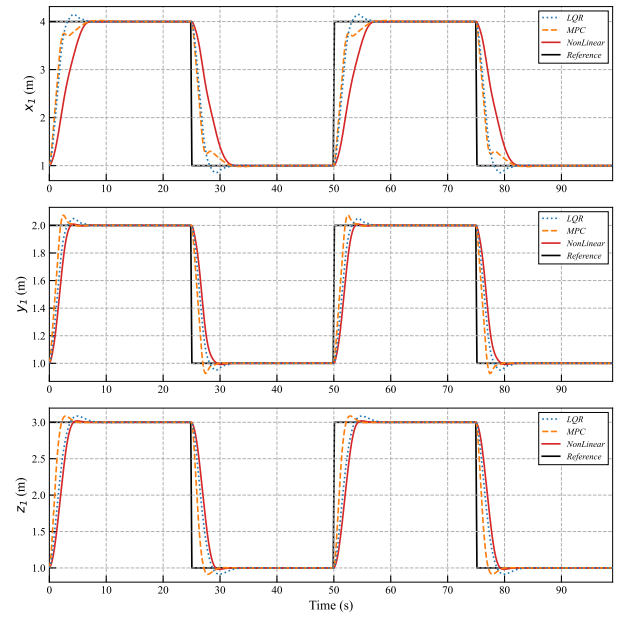
Numerical simulations are used to verify the effectiveness of the proposed control algorithm. The fourth-order Runge-Kutta method is used to solve the dynamic equation set. This study was conducted under the following Desktop configuration conditions:

1. The CPU is AMD Ryzen 7 7840HS 3.80 GHz.
2. The GPU is AMD Radeon 780M Graphics.
3. The RAM is 32GB.
4. The ROM is 1TB.
5. The OS is Windows 10 64bit.
6. The simulation software is MATLAB R2023b 64bit.

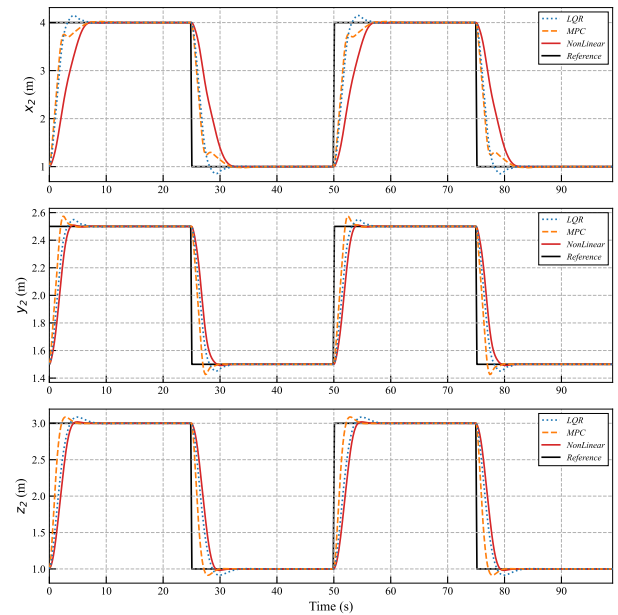
This section verifies the impact of variable rope length and load rotation on the model's response performance. Linear Quadratic Regulator (LQR) and Model Predictive Control (MPC) algorithms are used for comparison experiments, and the results of the proposed algorithm are compared with these traditional algorithms. Take the long pole as an example for hoisting the load.

Set the lifting load as a long pole. Before the simulation experiments, the model parameters must be determined. These are shown in Table 1.

In addition, the reference values for the UAVs positions and rope lengths are given in the form of a square wave with respect to time as the reference input to test the dynamic target tracking performance of the proposed control algorithm. To simulate actual engineering requirements, input saturation



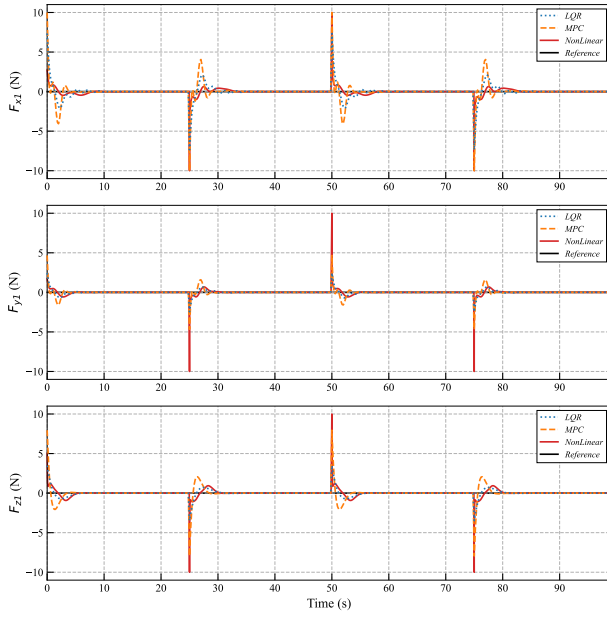
(a). Position Response of UAV 1.



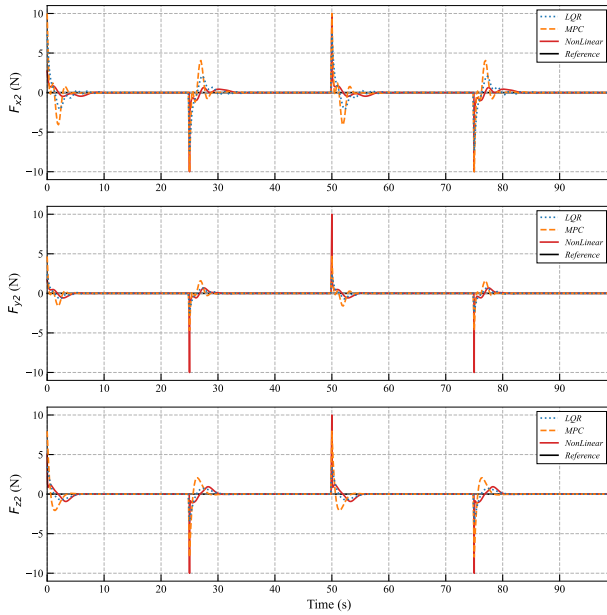
(b). Position Response of UAV 2.

FIGURE 4. Position response of UAVs.

constraints are set to $\pm 10N$ to limit the maximum output of the controller. In the numerical simulation experiments, parameter tuning methods were used to select control gains for the three algorithms. The control gains were systematically calibrated through an iterative optimization process that combines model-based sensitivity analysis with experimental validation. This hybrid approach fully utilizes prior knowledge of similar systems and real-time performance



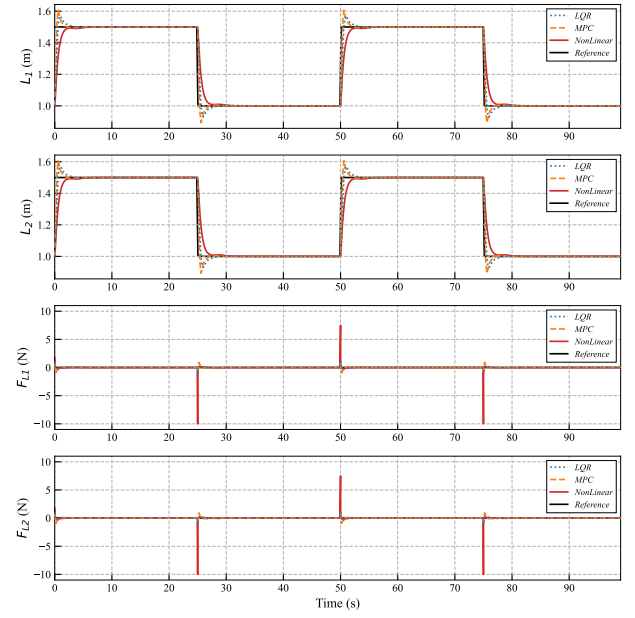
(a). Control Input of UAV 1.



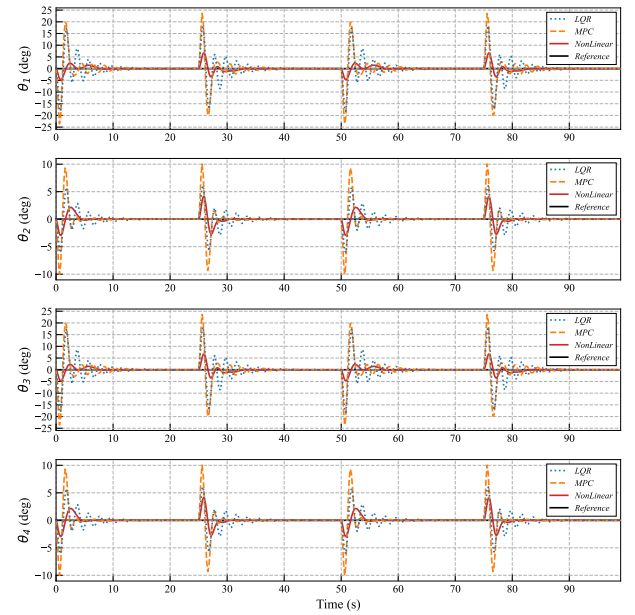
(b). Control Input of UAV 2.

FIGURE 5. Control input of UAVs.

feedback, ensuring adequate parameter configuration while maintaining dynamic stability. In addition, to ensure fairness, performance and the selected existing control designs were comprehensively considered. By adjusting parameters, the position tracking and control input errors of the three algorithms were set to the same magnitude. On this basis, the performance of suppressing the pendulum angle was compared. The relevant parameters of the controllers are shown as follows.



(a). Response of Rope Length and Control Input.



(b). Response of Swing Angle.

FIGURE 6. Response of rope length, Input, and Swing angle.

The control gain matrix K selected for the proposed Non-Linear algorithm is:

$$K = \begin{bmatrix} 0.6 & 2.4 & 0.6 & 1.4 & 0.6 & 1.2 \\ 0.6 & 2.4 & 0.6 & 1.4 & 0.6 & 1.2 \\ 1.2 & 2.6 & 1.2 & 2.6 & 1.2 & 2.6 \\ 0.6 & 2.4 & 0.6 & 1.4 & 0.6 & 1.2 \\ 0.6 & 2.4 & 0.6 & 1.4 & 0.6 & 1.2 \\ 1.2 & 2.6 & 1.2 & 2.6 & 1.2 & 2.6 \\ 1.2 & 1.2 & 1.2 & 1.2 & 1.2 & 0.9 \\ 1.2 & 1.2 & 1.2 & 1.2 & 1.2 & 0.9 \end{bmatrix} \quad (42)$$

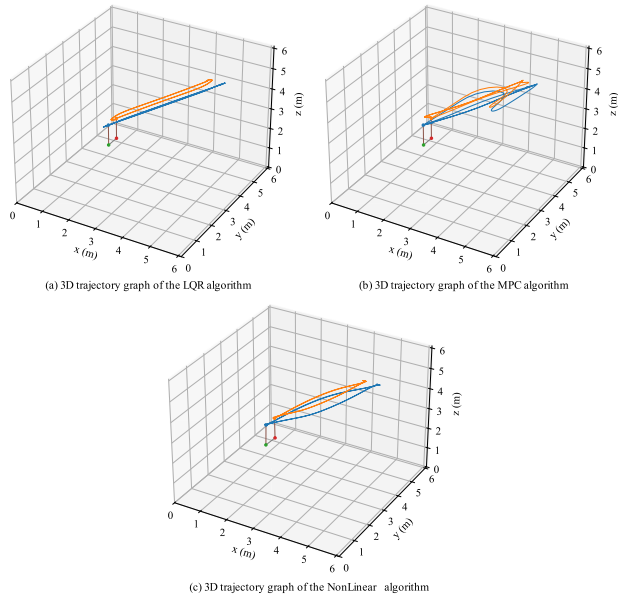


FIGURE 7. 3D Trajectory of UAVs lifting operation.

The matrix Q, R for the LQR algorithm is selected as follows:

$$\begin{aligned} Q &= \text{diag}([6, 6, 3, 6, 6, 3, 5, 5, 10, 10, 10, 10, \mathbf{0}_{1 \times 12}]), \\ R &= \text{diag}([\mathbf{1}_{1 \times 8}]) \end{aligned} \quad (43)$$

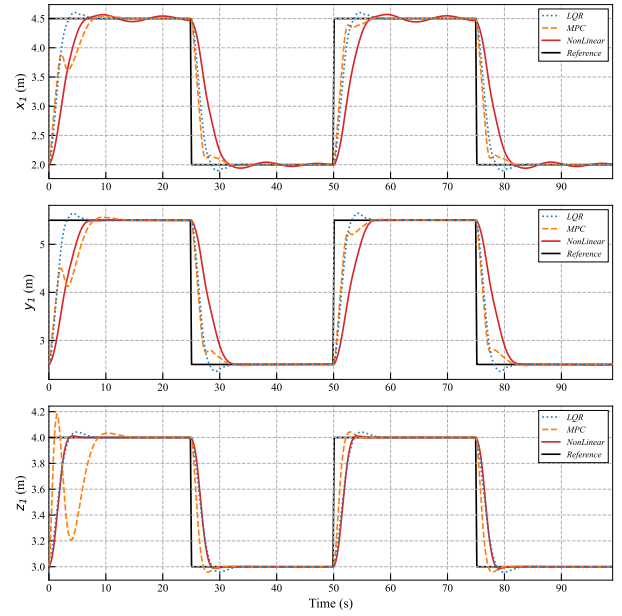
The matrix Q, R for the MPC algorithm is selected as follows:

$$\begin{aligned} Q &= \text{diag}([7, 7, 5, 7, 7, 5, 5, 5, 10, 10, 10, 10, \mathbf{0}_{1 \times 12}]), \\ R &= \text{diag}([\mathbf{1}_{1 \times 8}]) \end{aligned} \quad (44)$$

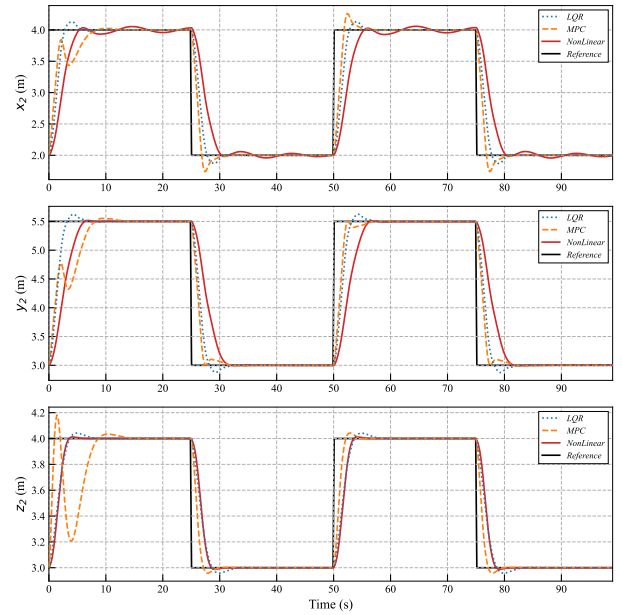
A. TRACKING CONTROL RESPONSE

This section designs a simulation comparison experiment for three control algorithms, where the LQR and MPC algorithms are primarily used for suppressing the swing of the load and positioning and navigating the UAVs. These are compared with the proposed control algorithm. Meanwhile, to further study the performance of the algorithms, four sets of control experiments were conducted under different scenarios, and the corresponding simulation trajectory figures were provided.

Figs. 3-6 show the simulation results for the first set of translation scenarios, with simulation parameters listed in Table 1. From the figures, it can be concluded that all three control algorithms can effectively and promptly track the position trajectory of the UAV, but the motion trajectory of the proposed algorithm is smoother. Although the nonlinear control algorithm exhibits a slightly longer adjustment time in x, y, z directions, approximately 6 s, compared to 4 s and 4 s for the LQR and MPC algorithms, respectively, the simulation results of the LQR and MPC algorithms indicate that they take longer to suppress the swing angle, with more



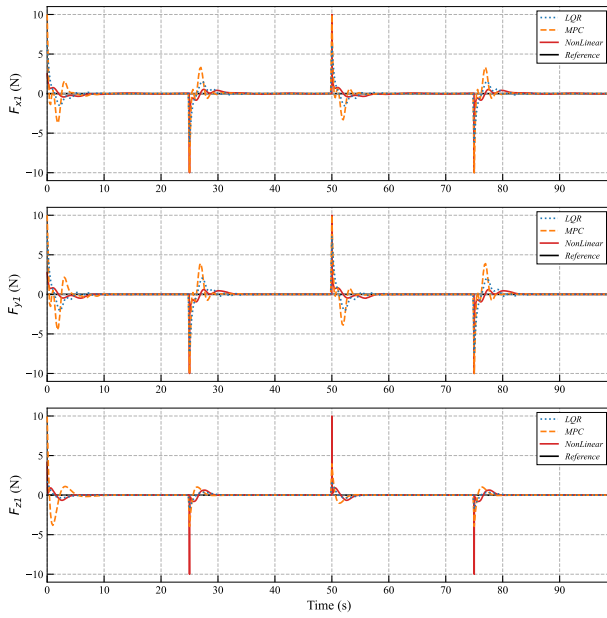
(a). Position Response of UAV 1.



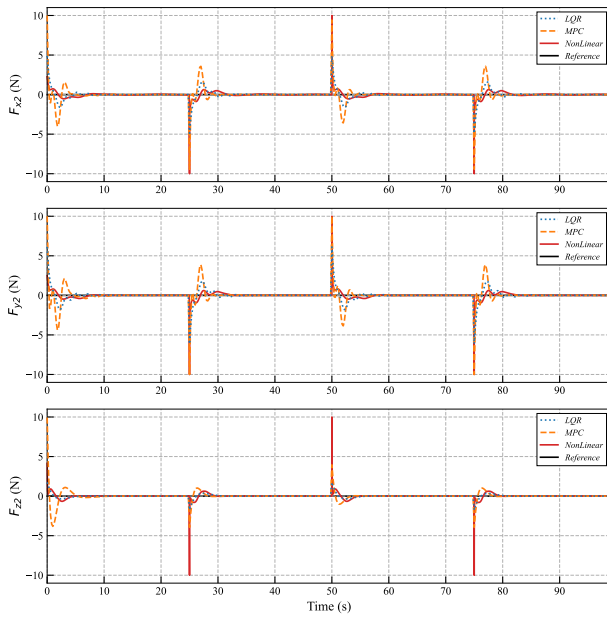
(b) Position Response of UAV 2.

FIGURE 8. Position response of UAVs.

oscillations and requiring 15 s to completely eliminate the swing. During this period, the amplitude of the swing is also relatively larger, about 18° and 24° , whereas the proposed control algorithm has a maximum swing angle of about 7° and can suppress the swing within 7 s after the trajectory change. Furthermore, the proposed control algorithm shows lower sensitivity to the rope length changes with square wave timing, while the LQR and MPC algorithms exhibit overshoot amounts of 0.1m and 0.05m. Figs. 5-6 compare the control



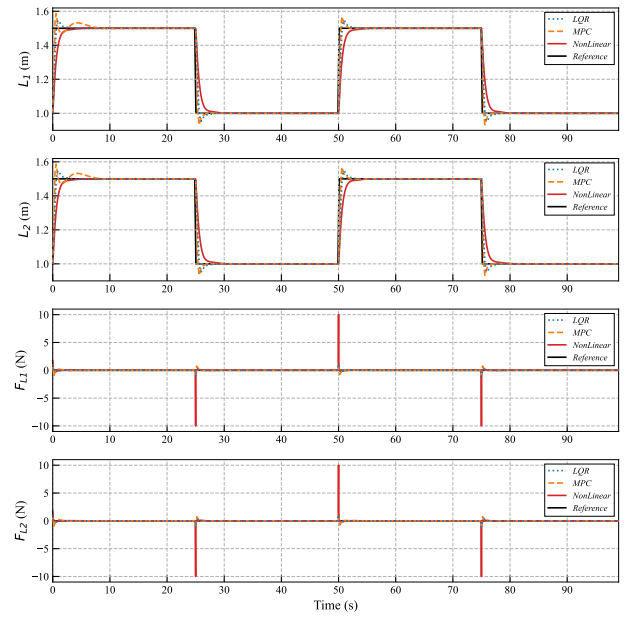
(a). Control Input of UAV 1.



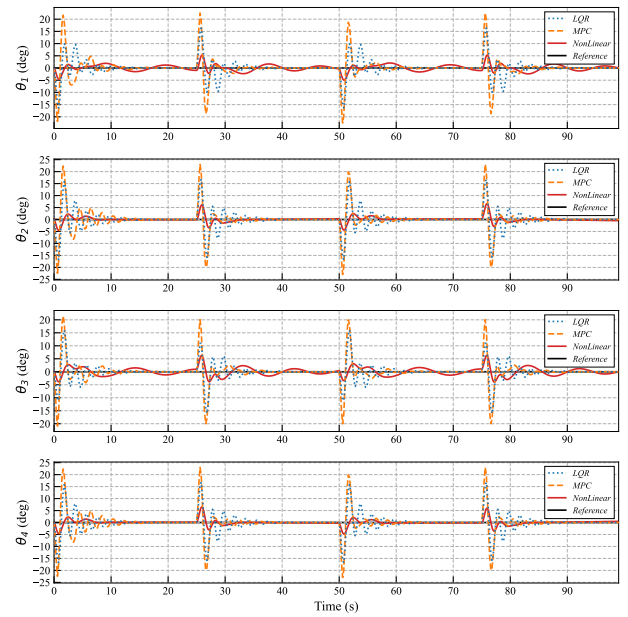
(b). Control Input of UAV 2.

FIGURE 9. Control input of UAVs.

inputs of the three algorithms. Under the input saturation limit of ± 10 N, the instantaneous input of the LQR and MPC algorithms during square wave switching is smaller than that of the nonlinear control algorithm, but both of these algorithms exhibit certain fluctuations afterward, with convergence times longer than the proposed algorithm. Overall, the nonlinear control algorithm demonstrates a smoother dynamic convergence process with smaller vibration amplitude, faster attenuation, and relatively fewer oscillations.



(a). Response of Rope Length and Control Input.



(b). Response of Swing Angle.

FIGURE 10. Response of rope length, Input, and Swing angle.

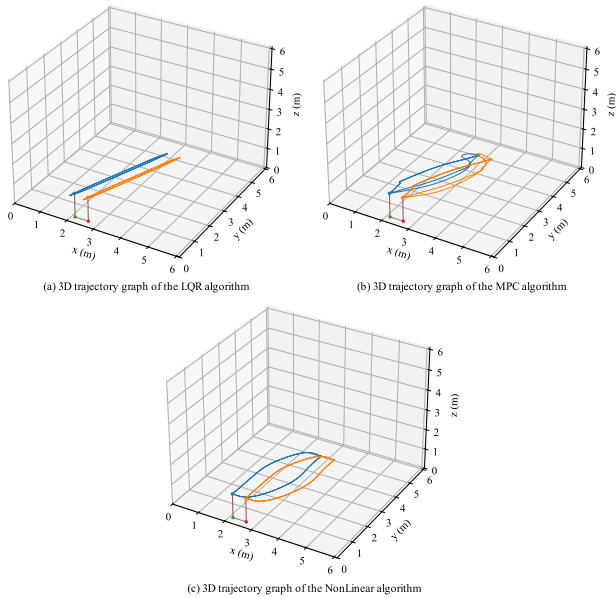
Next, considering the rotational scenario, this is set as the second group of experiments. With the other parameters remaining unchanged, the initial and target positions of the UAV are set as follows:

$$\begin{aligned} p_1 &= (2 \text{ m}, 2.5 \text{ m}, 3 \text{ m}), p_2 = (2 \text{ m}, 3 \text{ m}, 3 \text{ m}) \\ p_{1d} &= (4.5 \text{ m}, 5.5 \text{ m}, 4 \text{ m}), p_{2d} = (4 \text{ m}, 5.5 \text{ m}, 4 \text{ m}) \end{aligned} \quad (45)$$

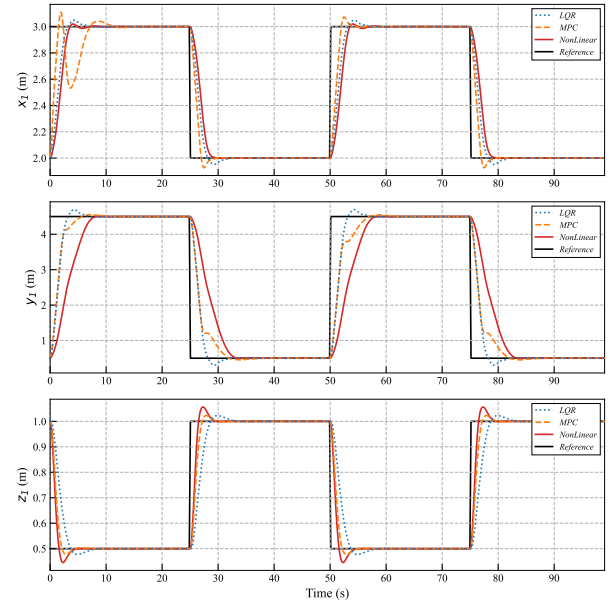
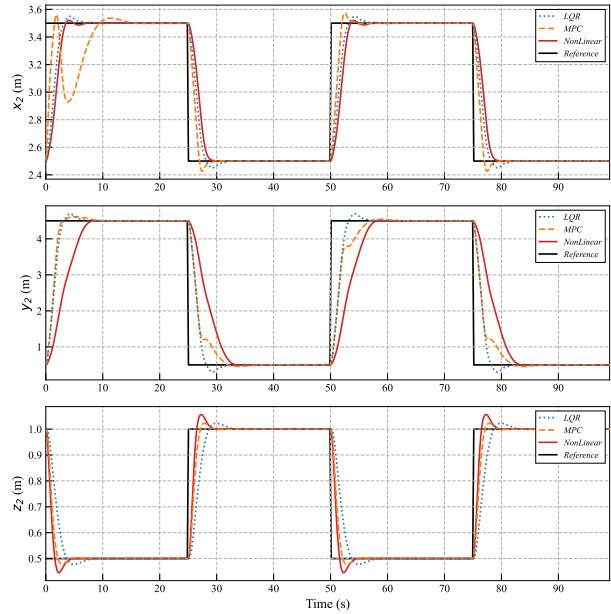
In this set of simulation experiments, as shown in Figs. 7-10, the proposed NonLinear algorithm shows high adherence

TABLE 2. Performance comparison.

Symbol	Linear Quadratic Regulator (LQR)	Model Predictive Control (MPC)	Nonlinear Control Algorithm
e_{1rms}	[0.55, 0.65, 0.23, 0.06]	[0.51, 0.65, 0.21, 0.06]	[1.07, 1.31, 0.38, 0.13]
e_{2rms}	[0.43, 0.54, 0.23, 0.06]	[0.38, 0.52, 0.21, 0.06]	[0.84, 1.07, 0.38, 0.13]
u_{1rms}	[0.81, 1.00, 0.25, 0.12]	[1.20, 1.29, 0.75, 0.12]	[0.95, 0.98, 1.00, 0.79]
u_{2rms}	[0.69, 0.83, 0.25, 0.12]	[1.18, 1.25, 0.75, 0.12]	[0.93, 0.95, 1.00, 0.79]
θ_{max}	[16.36, 16.69, 15.99, 16.69]	[22.54, 22.98, 21.47, 22.98]	[5.29, 6.60, 6.28, 6.49]
θ_{rms}	[3.41, 3.87, 3.32, 3.87]	[4.48, 4.70, 4.30, 4.70]	[1.60, 1.50, 1.83, 1.51]

**FIGURE 11.** 3D Trajectory of UAVs lifting operation.

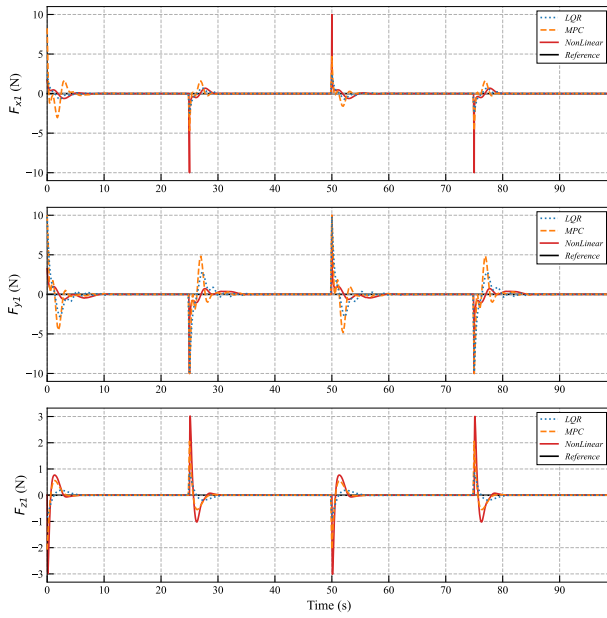
to tracking targets across displacement and swing angle parameters in each dimension, with significant advantages in tracking accuracy. Regarding convergence and response, during square wave signal state transitions, the NonLinear algorithm quickly converges to the reference trajectory, and it responds swiftly and adjusts the swing angle to the reference value. In contrast, the LQR and MPC algorithms exhibit relatively large swing angle overshoot, with maximum swing angles reaching 15° and 20° , whereas the proposed algorithm achieves only 5° , which is 33% and 25% of the former values. Moreover, the contrast algorithms show slow convergence and response, with noticeable oscillations during the process; although the MPC algorithm performs better in convergence than the LQR algorithm, it still falls short of the proposed NonLinear algorithm in terms of convergence speed and response stability.

**(a).** Position Response of UAV 1.**(b).** Position Response of UAV 2.**FIGURE 12.** Position response of UAVs.

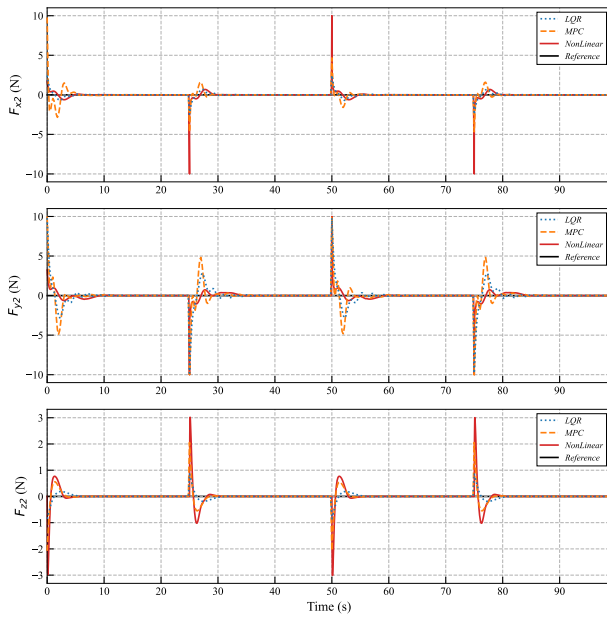
B. QUANTITATIVE PERFORMANCE COMPARISON

To further evaluate the performance of the proposed control algorithm, it is compared quantitatively with the response results of the LQR and MPC algorithms. The maximum swing angle $\theta_{max} = |\theta|_{max}$ and the root mean square (RMS) values of the average load swing angle, tracking error, and control input are used as evaluation metrics θ_{rms} , e_{rms} , u_{rms} . The relevant definitions of the performance metrics are shown in Appendix.

Table 2 shows the performance metrics generated through the aforementioned definitions, with the simulated results



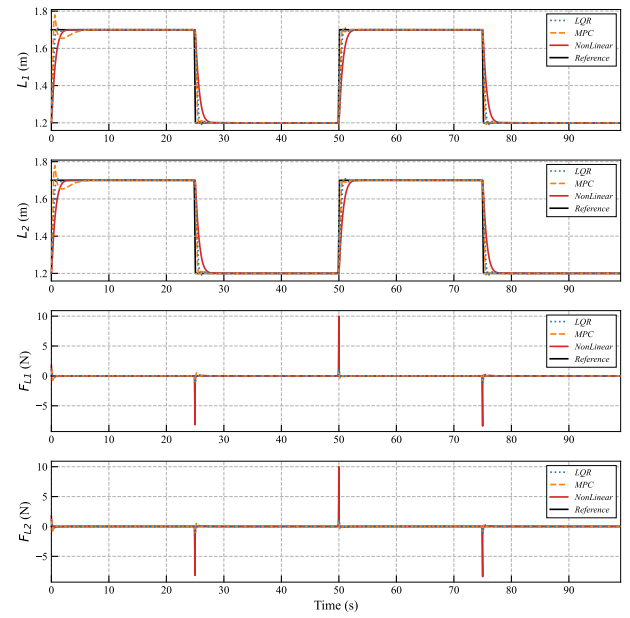
(a). Control Input of UAV 1.



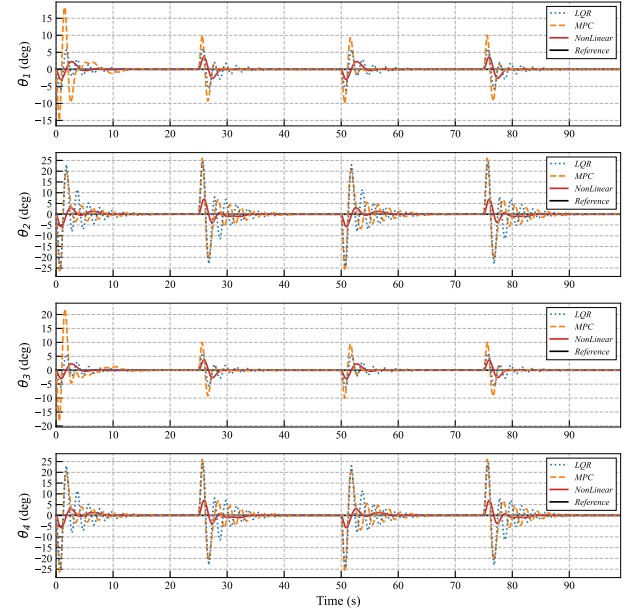
(b). Control Input of UAV 2.

FIGURE 13. Control Input of UAVs.

consistent with the Group 2. The simulation results are consistent with the examples. With a fair comparison as the premise, the simulation parameters were set to maintain consistency. All three algorithms are required to provide approximately the same position tracking performance and require almost the same control input, as shown in the results of e_{rms} and u_{rms} in Table 2. By comparing the responses of the three control algorithms, it can be seen that all three algorithms can reach the desired position with



(a). Response of Rope Length and Control Input.



(b). Response of Swing Angle.

FIGURE 14. Response of rope length, Input, and Swing angle.

similar average tracking errors and control inputs. However, as shown in Figs. 7-10 and Table 2, although all three algorithms can achieve the desired target, the proposed algorithm has a smaller maximum load swing angle θ_{max} , improving by 67.67%, 60.46%, 60.73%, and 61.11% over the LQR algorithm, and by 76.53%, 71.28%, 70.75%, and 71.76% over the MPC algorithm respectively. The results of θ_{rms} in 2 prove that the proposed algorithm effectively suppresses the load swing.

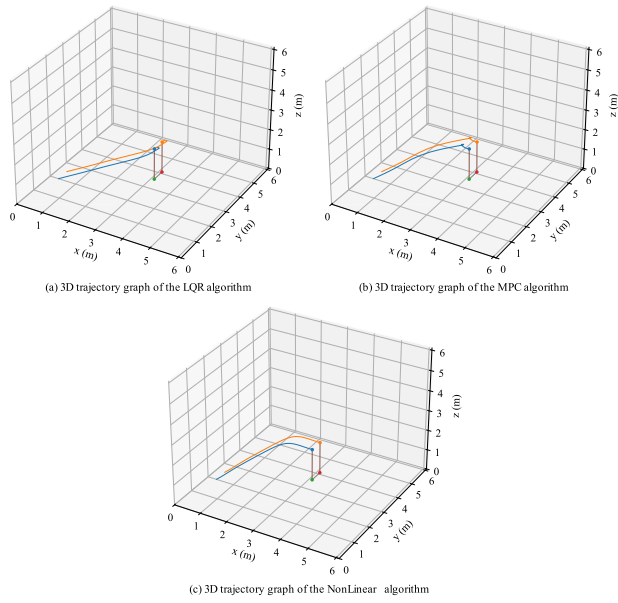


FIGURE 15. 3D Trajectory of drone lifting operation.

C. ROBUSTNESS TEST

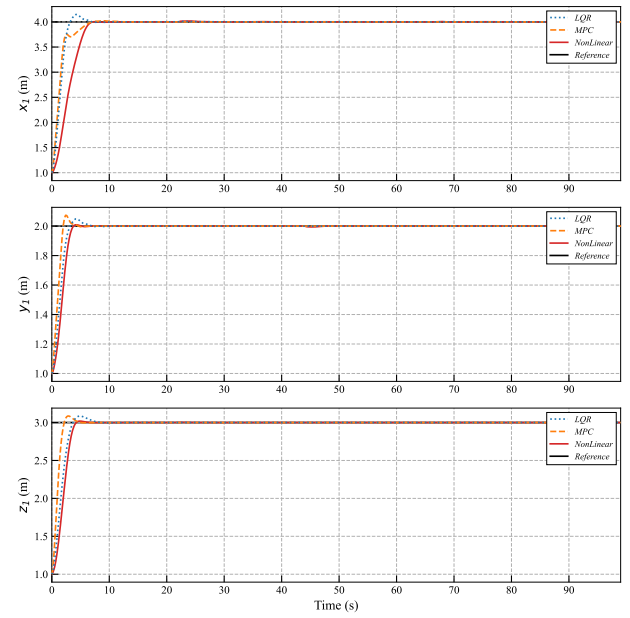
To verify the robustness of the proposed algorithm under uncertain parameters, the some parameters are changed as follows:

$$\begin{aligned} m_3 &= 0.4 \text{ kg}, L = (1.2 \text{ m}, 1.2 \text{ m}), L_d = (1.7 \text{ m}, 1.7 \text{ m}) \\ I_{xx} &= 0.3 \text{ kg} \cdot \text{m}^2, I_{yy} = 0.3 \text{ kg} \cdot \text{m}^2, I_{zz} = 0.3 \text{ kg} \cdot \text{m}^2 \\ p_1 &= (2 \text{ m}, 0.5 \text{ m}, 1 \text{ m}), p_2 = (2.5 \text{ m}, 0.5 \text{ m}, 1 \text{ m}) \\ p_{1d} &= (3 \text{ m}, 4.5 \text{ m}, 0.5 \text{ m}), p_{2d} = (3.5 \text{ m}, 4.5 \text{ m}, 0.5 \text{ m}) \end{aligned} \quad (46)$$

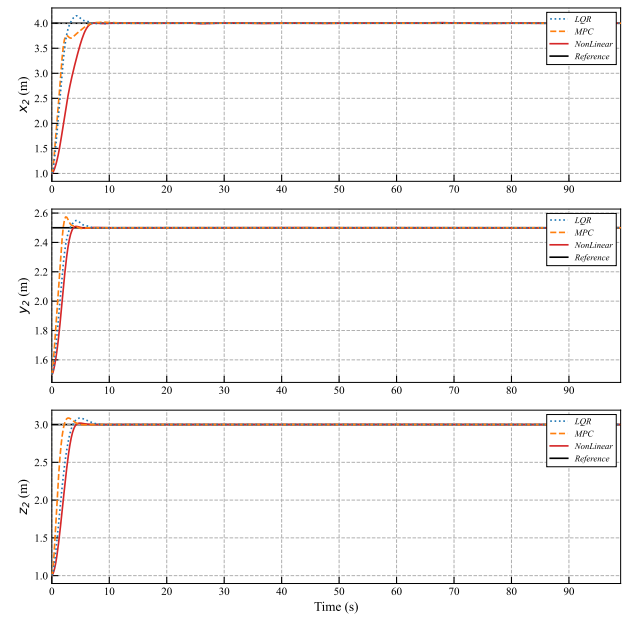
The results of the system responses are shown in Figs. 11-14. It can be seen that all three control methods can achieve the positioning target of the UAVs. The changes in the parameters have no significant impact on the displacement and rope length tracking of the proposed algorithm. However, the tracking response of the LQR and MPC algorithms exhibits a certain amount of overshoot, especially with the MPC algorithm showing a severe fluctuation of a 0.6 m drop during the initial stage of tracking in the x direction, with a settling time of up to 12 s. Moreover, in the cable length tracking, the overshoot of the MPC algorithm is 0.1 m, which is also significantly higher than that of the proposed algorithm.

As can be seen from Fig. 14, while all three methods are capable of achieving the goal of damping the swing, the nonlinear control algorithm proposed in this paper still exhibits better performance in suppressing the angle of the swing. In contrast, the other two control methods exhibit larger oscillations and longer convergence times. After comprehensive comparison, the control algorithm proposed in this paper can better handle parameter uncertainties.

To simulate the effects of wind and other external disturbances on the system response, the fourth experiment is set up based on the parameters in Table 1. First, the multi-UAV



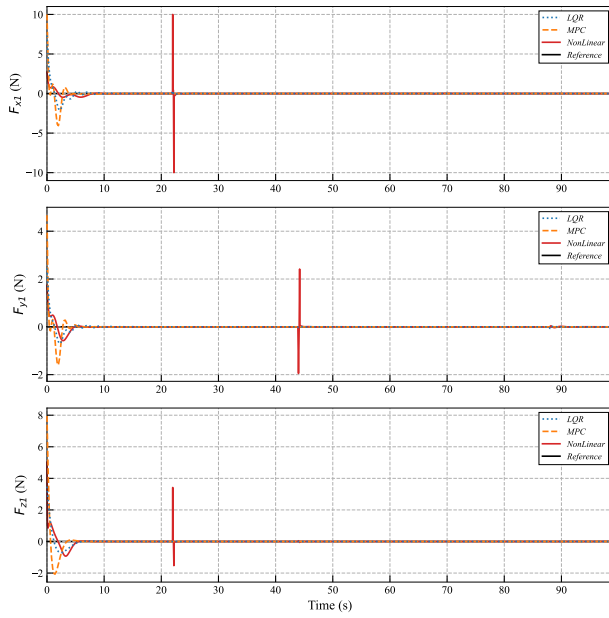
(a). Position Response of UAV 1.



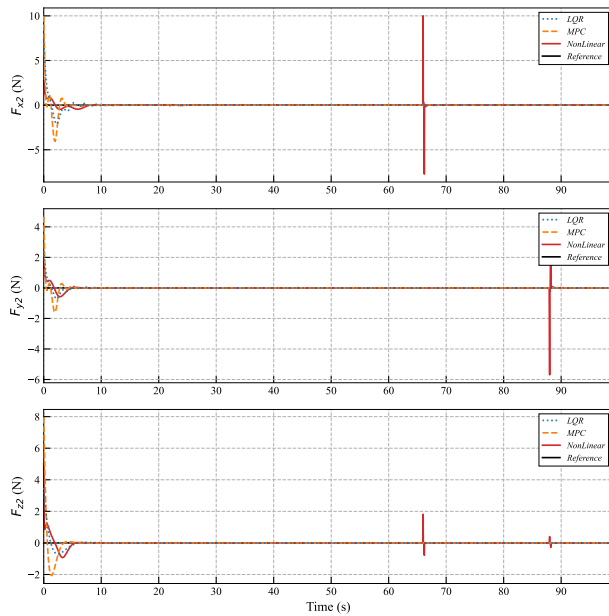
(b). Position Response of UAV 2.

FIGURE 16. Position response of UAVs.

lifting system was first made to track a constant reference position. The closed-loop system quickly reached stability. Subsequently, at 22s, 44s, 66s, and 88s, a pulse disturbance with an amplitude of 10° , -2° , 5° , and -7° were applied to the swing angle $\theta = [\theta_1, \theta_2, \theta_3, \theta_4]^T$ of the hoisting system under the three controllers that were in operation. The response results are shown in Figs. 15-18. When external disturbances interfered, the offset oscillations of the three algorithms in the position tracking were very small and could be ignored. Importantly, the algorithm proposed in this paper,



(a). Control Input of UAV 1.

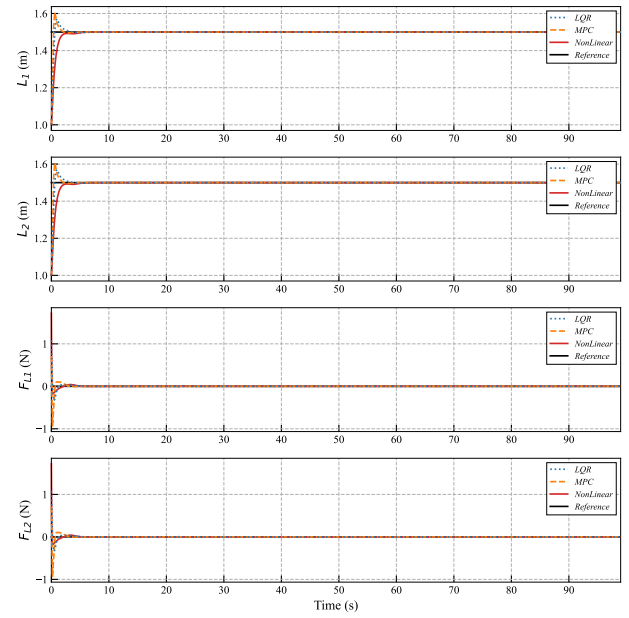


(b). Control Input of UAV 2.

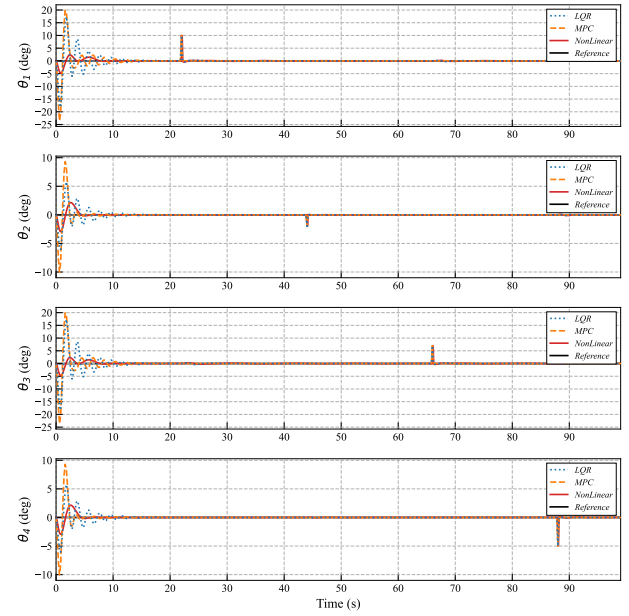
FIGURE 17. Control input of UAVs.

while starting with a smaller initial oscillation, was still able to quickly suppress the oscillation of the load swing angle caused by disturbances, similar to the other two algorithms. As shown in Fig. 18, the load swing angle approached 0° in approximately 2 s. This indicated that under the same conditions, the nonlinear control algorithm not only demonstrated excellent performance but also exhibited good resistance to external disturbances.

In summary, the simulation results show that the proposed control algorithm can effectively suppress the load swing



(a). Response of Rope Length and Control Input.



(b). Response of Swing Angle.

FIGURE 18. Response of rope length, Input, and Swing angle.

even when the system model considers the variable cable length and introduces the coupling effect of the load attitude. The algorithm can still perform well in controlling the load swing when the hoisting system is disturbed, based on the nonlinear coupling relationship between the UAV displacements, cable length, and load swing angle.

VI. CONCLUSION

This paper addresses the control challenges of a multi-UAV cooperative transportation system by proposing a Lyapunov

nonlinear control strategy based on passive energy analysis. It overcomes the traditional fixed rope length assumption and innovatively introduces winch driving force as a fourth-dimensional control variable to construct a high-dimensional dynamic model that includes time-varying rope length, load attitude coupling, and multi-UAV kinematic constraints. By constructing an analytical auxiliary error function, it establishes a nonlinear coupling relationship between UAV displacement, rope length, and swing angle, and designs a nonlinear error feedback mechanism to achieve active compensation of load swing, thereby enhancing system control accuracy. Simulation results indicate that, compared to the LQR and MPC baseline algorithms, this control strategy significantly reduces load swing amplitude while maintaining the same level of position tracking accuracy, with the maximum swing angle reduced by approximately 60%-76% and convergence time shortened by about 50%, and the smoothness of control input increased by over 30%. Furthermore, under conditions of parameter uncertainty and multi-level wind disturbances, the proposed method maintained stable tracking, whereas the LQR and MPC algorithms exhibited significant overshoot and oscillations under disturbance conditions, further validating the robustness of the proposed control strategy.

However, this paper still lacks in the aspects of method expansion and practical validation. In future research, the method will be further expanded, considering the integration of deep reinforcement learning to enhance system adaptability. A physical experimental platform will be built to verify the reliability of the algorithm in real-world complex environments.

APPENDIX

DEFINITION OF THE PERFORMANCE METRICS

The performance metrics are defined as follows:

$$\begin{aligned} e_{1rms} &= [e_{x1rms}, e_{y1rms}, e_{z1rms}, e_{L1rms}] \\ e_{2rms} &= [e_{x2rms}, e_{y2rms}, e_{z2rms}, e_{L2rms}] \\ u_{1rms} &= [u_{1rms}, u_{2rms}, u_{3rms}, u_{7rms}] \\ u_{2rms} &= [u_{4rms}, u_{5rms}, u_{6rms}, u_{8rms}] \\ \theta_{\max} &= [\theta_{1\max}, \theta_{2\max}, \theta_{3\max}, \theta_{4\max}] \\ \theta_{rms} &= [\theta_{1rms}, \theta_{2rms}, \theta_{3rms}, \theta_{4rms}] \end{aligned} \quad (A.1)$$

where

$$\begin{aligned} e_{rms} &= \sqrt{\frac{1}{\Delta T} \int_{T_{down}}^{T_{up}} e^2(t) dt} \\ u_{rms} &= \sqrt{\frac{1}{\Delta T} \int_{T_{down}}^{T_{up}} u^2(t) dt} \\ \theta_{rms} &= \sqrt{\frac{1}{\Delta T} \int_{T_{down}}^{T_{up}} \theta^2(t) dt} \\ \Delta T &= T_{up} - T_{down} \end{aligned} \quad (A.2)$$

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